

1. Father of Cloud Computing is:
 - A. Tim Berners-Lee
 - B. J.C.R. Licklider
 - C. Bill Gates
 - D. Larry Page
2. MARR stands for:
 - A. Minimum Acceptable Rate of Return
 - B. Maximum Annual Rate of Return
 - C. Marginal Average Rate of Return
 - D. Minimum Annual Revenue Rate
3. Ant Colony Optimization algorithm was proposed by:
 - A. John Holland
 - B. Dorigo
 - C. Marvin Minsky
 - D. Alan Turing
4. For an LCR circuit with $i=20\sin(157t)$, frequency is:
 - A. 25 Hz
 - B. 50 Hz
 - C. 157 Hz
 - D. 100 Hz
5. (7,4) Hamming code can:
 - A. Detect 1 error
 - B. Correct 1 error
 - C. Correct 2 errors
 - D. Detect 2 errors only
6. Auditor of Nepal Engineering Council is appointed by:
 - A. NEC Board
 - B. Government of Nepal
 - C. Prime Minister
 - D. Engineer's Association
7. Knowledge-based system is a part of:
 - A. Database system
 - B. Artificial Intelligence
 - C. Operating System
 - D. Computer Network
8. An example of intelligent agent is:
 - A. Calculator
 - B. Compiler
 - C. Self-driving car
 - D. Printer
9. Average time complexity of Shell Sort is:
 - A. $O(n)$
 - B. $O(n \log n)$
 - C. $O(n^2)$
 - D. Depends on gap sequence
10. Ambient light means:
 - A. Artificial light
 - B. Reflected light
 - C. Surrounding natural light
 - D. Focused beam
11. View where observer is above object looking down is:
 - A. Front view
 - B. Side view
 - C. Top view
 - D. Bottom view
12. IPv4 address size is:
 - A. 16 bits
 - B. 32 bits
 - C. 64 bits
 - D. 128 bits
13. Input part of biological neural network is:
 - A. Axon
 - B. Dendrite

- C. Synapse
D. Cell body
14. Output of D Flip-Flop depends on:
A. Clock only
B. Previous output
C. D input at clock edge
D. Reset only
15. Gate used to check odd parity is:
A. AND
B. OR
C. NAND
D. XOR
16. Least probability of error occurs in:
A. ASK
B. FSK
C. PSK
D. PAM
17. Bridge operates at:
A. Physical layer
B. Network layer
C. Transport layer
D. Data Link layer
18. Data about data is called:
A. Raw data
B. Information
C. Metadata
D. Knowledge
19. "Data is data" represents:
A. Metadata
B. Raw data
C. Information
D. Knowledge
20. Which is NOT a unit in Ohm's law?
A. Volt
B. Ampere
C. Ohm
D. Farad
21. Reactive power is expressed in:
A. Watt
B. Volt
C. VAR
D. Joule
22. Number of flags in 8085 microprocessor is:
A. 3
B. 4
C. 5
D. 6
23. Number of flip-flops in 8085 flag register is:
A. 3
B. 4
C. 5
D. 8
24. Dividing memory into equal size blocks is called:
A. Fragmentation
B. Paging
C. Segmentation
D. Swapping
25. Which sorting algorithm does NOT use recursion?
A. Quick sort
B. Merge sort
C. Heap sort
D. Tree sort
26. Object-oriented design has how many layers?
A. 2
B. 3

- C. 4
D. 5
27. Software quality is checked by:
A. Compilation
B. Debugging
C. Verification & Validation
D. Coding
28. C++ is a:
A. Procedural language
B. Assembly language
C. Object-oriented language
D. Markup language
29. Which is NOT an HTML tag?
A. <p>
B. <body>
C. <text>
D. <div>
30. Encapsulation means:
A. Inheritance
B. Polymorphism
C. Data hiding
D. Abstraction
31. Wien Bridge is used as:
A. Amplifier
B. Oscillator
C. Rectifier
D. Filter
32. POP3 is used for:
A. Sending email
B. Receiving email
C. Browsing web
D. Alert message
33. Which image format is NOT lossless?
A. GIF
B. PNG
C. BMP
D. JPG
34. Which transformation does NOT change shape?
A. Scaling
B. Rotation
C. Translation
D. Both B and C
35. Color of light depends on:
A. Speed
B. Amplitude
C. Wavelength
D. Phase
36. For a graph with n nodes, maximum edges is:
A. n
B. $n+1$
C. $n-1$
D. $n(n-1)/2$
37. 2's complement of a 7-bit number gives:
A. Same bits
B. Sign inversion
C. One's complement
D. No change
38. When temperature increases in silicon diode:
A. Resistance increases
B. Current decreases
C. Reverse saturation current increases
D. Voltage remains constant
39. Bandwidth is measured in:
A. Volts
B. Hertz

- C. Seconds
D. Watts
40. Serial communication transmits data:
A. Bit by bit
B. Byte by byte
C. Block by block
D. Word by word
41. In 8086, AL register is:
A. 16-bit
B. 8-bit
C. 32-bit
D. 4-bit
42. Unit of sound frequency is:
A. Decibel
B. Watt
C. Hertz
D. Joule
43. European E2 standard frequency is:
A. 8.448 MHz
B. 8.844 MHz
C. 8.570 MHz
D. 9.000 MHz
44. Empty class in C++ occupies:
A. 0 byte
B. 1 byte
C. 2 bytes
D. Undefined
45. Socket style for Windows is called:
A. wsock
B. winsock
C. sockwi
D. sins
46. UDP is:
A. Connection-oriented & reliable
B. Connectionless & reliable
C. Connectionless & unreliable
D. Connection-oriented & unreliable
47. Which is NOT a polygon mesh?
A. Triangle
B. Quadrilateral
C. Triangle strip
D. Bezier surface
48. Bezier curve blending functions sum equals:
A. 0
B. 1
C. 2
D. n
49. Number of pins in ATM switch is:
A. 48
B. 52
C. 54
D. 56
50. Sutherland-Cohen algorithm is used for:
A. Filling
B. Line clipping
C. Polygon shading
D. Projection
51. Number of articles proposed by NEC Act:
A. 6
B. 7
C. 8
D. 9
52. To convert 4NF to 5NF, decomposition must be:
A. Lossy
B. Lossless

- C. Partial
D. Random
53. Which is NOT a behavioral model?
- A. Data flow
B. State transition
C. Use case
D. Structure
54. R in AC circuit represents:
- A. Reactance
B. Impedance
C. Resistance
D. Resonance
55. Transparent device refers to:
- A. JK flip-flop
B. SR flip-flop
C. D latch
D. T flip-flop
56. 81-90-4405-4020 is an example of:
- A. IPv4 address
B. Phone number
C. MAC address
D. ISBN

2. B - Licklider proposed global computer networking, the base idea of cloud computing.
3. A - It is the minimum return required for an investment to be acceptable.
4. B - Marco Dorigo developed ACO based on ant behavior.
5. B - Since $\omega=2\pi f$, $f=157/2\pi\approx 50$ Hz.
6. B - Hamming (7,4) can correct one-bit error.
7. B - NEC Act authorizes the Government to appoint the auditor.
8. B - Knowledge-based systems are core applications of AI.
9. C - It senses environment and acts autonomously to achieve goals.
10. D - Shell sort performance varies with chosen gap sequence.
11. C - Ambient light is uniform background light present everywhere.
12. C - Top view shows object from above.
13. B - IPv4 uses 32-bit addressing.
14. B - Dendrites receive signals from other neurons.
15. C - D flip-flop stores input value at clock edge.
16. D - XOR outputs 1 when number of 1s is odd.
17. C - PSK is most noise-resistant among given options.
18. D - Bridge uses MAC addresses at Layer 2.
19. C - Metadata describes properties of data.
20. B - Unprocessed facts are called raw data.
21. D - Farad is unit of capacitance, not Ohm's law.
22. C - Reactive power unit is Volt-Ampere Reactive (VAR).
23. C - 8085 has five flags (S, Z, AC, P, CY).
24. C - Only 5 bits are implemented using flip-flops.
25. B - Paging divides memory into fixed-size blocks.
26. C - Heap sort is implemented iteratively.
27. B - Typical OOD uses UI, logic, and data layers.
28. C - Verification checks correctness, validation checks user needs.
29. C - C++ supports classes, objects, and OOP features.
30. C - `<text>` is not a standard HTML tag.
31. C - Encapsulation hides internal data using classes.
32. B - Wien bridge generates sinusoidal signals.
33. B - POP3 retrieves emails from mail server.
34. D - JPG uses lossy compression.
35. D - Rotation and translation preserve shape.
36. C - Different wavelengths correspond to different colors.
37. D - Each node can connect to all others once.

- 38. B - 2's complement represents negative of number.
- 39. C - Heat increases minority carrier generation.
- 40. B - Bandwidth is frequency range.
- 41. A - Serial transmission sends one bit at a time.
- 42. B - AL is lower 8 bits of AX register.
- 43. C - Frequency is measured in Hertz.
- 44. A - E2 standard data rate is 8.448 Mbps.
- 45. B - C++ allocates 1 byte for unique address.
- 46. B - Winsock is Windows socket API.
- 47. C - UDP does not guarantee delivery.
- 48. D - Bezier surface is mathematically curved.
- 49. B - Blending functions satisfy partition of unity.
- 50. D - Some ATM hardware uses 56-pin configuration.
- 51. B - It clips lines within a viewport.
- 52. C - NEC Act is structured around 8 articles.
- 53. B - Lossless join preserves original data.
- 54. D - Structure represents static system design.
- 55. C - R is symbol for resistance.
- 56. C - D latch passes input when enabled.
- 57. B